

NAME: CHINWE EMMANUELLA IDIGO

EMAIL: idigochinwe2001@gmail.com

SQL CAPSTONE PROJECT

Q1: What are the business's profit drivers?

SELECT

```
p."Product_Category",  
SUM(sa."Units" * p."Product_Price") AS total_revenue,  
SUM(sa."Units" * p."Product_Cost") AS total_cost,  
SUM(sa."Units" * p."Product_Price" - sa."Units" * p."Product_Cost") AS total_profit
```

FROM sales sa

JOIN products p ON sa."Product_ID" = p."Product_ID"

GROUP BY p."Product_Category"

ORDER BY total_profit DESC;

Product_Category	Total_Revenue	Total_Cost	Total_Profit
Toys	\$5,093,241.00	\$4,013,714.00	\$1,079,527.00
Electronics	\$2,246,771.25	\$1,245,334.25	\$1,001,437.00
Art & Crafts	\$2,705,364.26	\$1,952,010.26	\$753,354.00
Games	\$2,226,836.27	\$1,552,843.27	\$673,993.00
Sports & Outdoors	\$2,172,359.57	\$1,666,641.57	\$505,718.00

Insight: The Toys category is Maven Toys' biggest profit driver, generating \$1,079,527.00 in total profit. It is closely followed by Electronics with \$1,001,437.00. Analysis: Although Toys has the highest revenue, Electronics has a better profit margin of 44.6% compared to Toys' 21.2%. This means Electronics is more efficient at turning sales into profit. Recommendation: Maven Toys should continue investing in Toys for high volume, but should strategically expand the Electronics category due to its superior margin. The Sports & Outdoors category has the lowest profit and requires a review of pricing or costs.

Q1b: Profit drivers by store location

SELECT

s."Store_City",

p."Product_Category",

SUM(sa."Units" * p."Product_Price" - sa."Units" * p."Product_Cost") AS total_profit

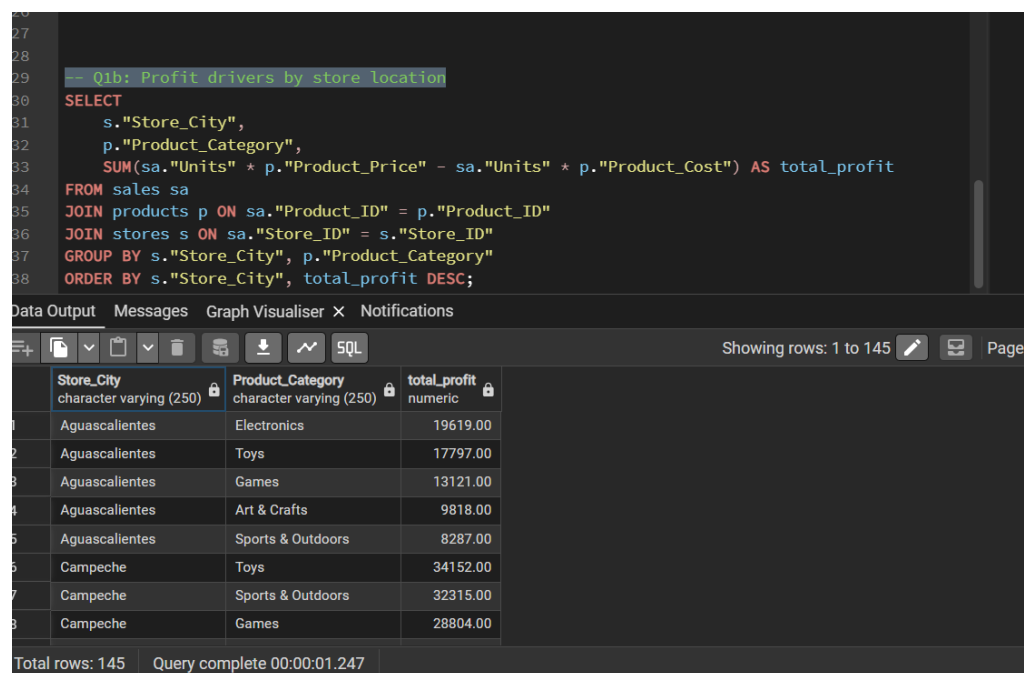
FROM sales sa

JOIN products p ON sa."Product_ID" = p."Product_ID"

JOIN stores s ON sa."Store_ID" = s."Store_ID"

GROUP BY s."Store_City", p."Product_Category"

ORDER BY s."Store_City", total_profit DESC;

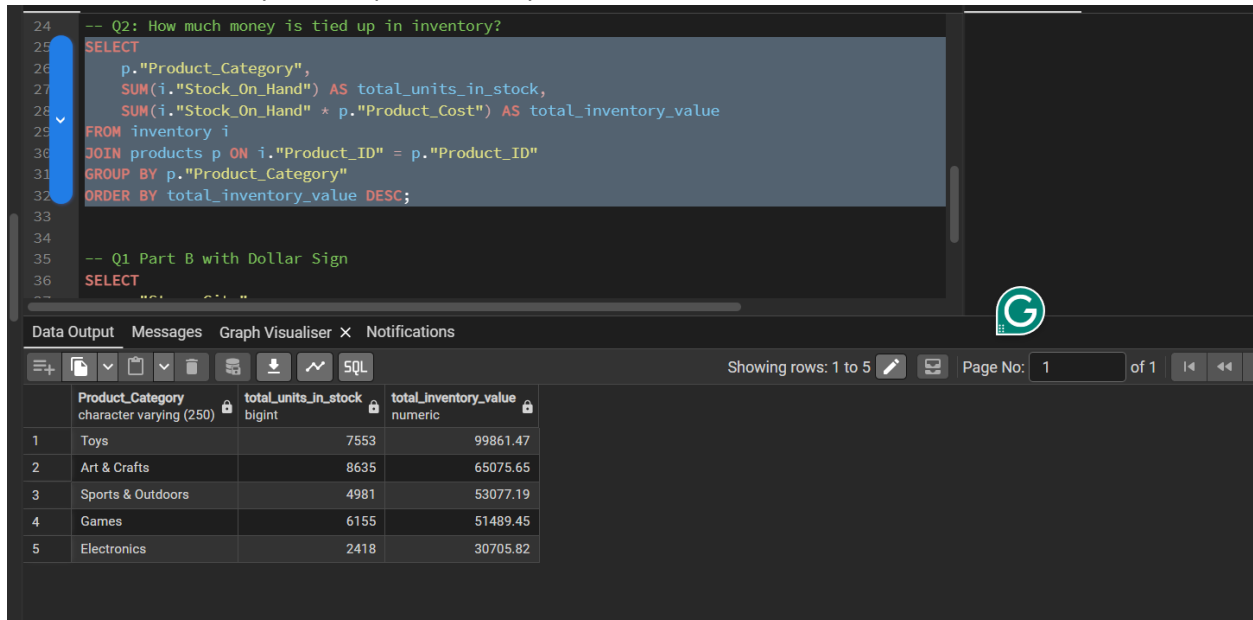


The screenshot shows a SQL IDE interface. The top pane displays the SQL query for Q1b: Profit drivers by store location. The bottom pane shows the results of the query, which are 145 rows. The results are displayed in a table with three columns: Store_City, Product_Category, and total_profit. The first few rows are visible, showing data for Aguascalientes and Campeche.

Store_City	Product_Category	total_profit
Aguascalientes	Electronics	19619.00
Aguascalientes	Toys	17797.00
Aguascalientes	Games	13121.00
Aguascalientes	Art & Crafts	9818.00
Aguascalientes	Sports & Outdoors	8287.00
Campeche	Toys	34152.00
Campeche	Sports & Outdoors	32315.00
Campeche	Games	28804.00

Insight: Profit drivers vary by store location. For example, Electronics is the top profit driver in Aguascalientes with \$19,619.00, while Toys is the top driver in Campeche with \$34,152.00. This pattern continues across the 145 city-category combinations.

Q2. How much money is tied up in inventory



The screenshot shows a SQL IDE interface. The top pane contains a SQL query for Q2: "How much money is tied up in inventory?". The query selects product categories, total units in stock, and total inventory value, ordered by total inventory value in descending order. The bottom pane shows the "Data Output" tab with a table of results.

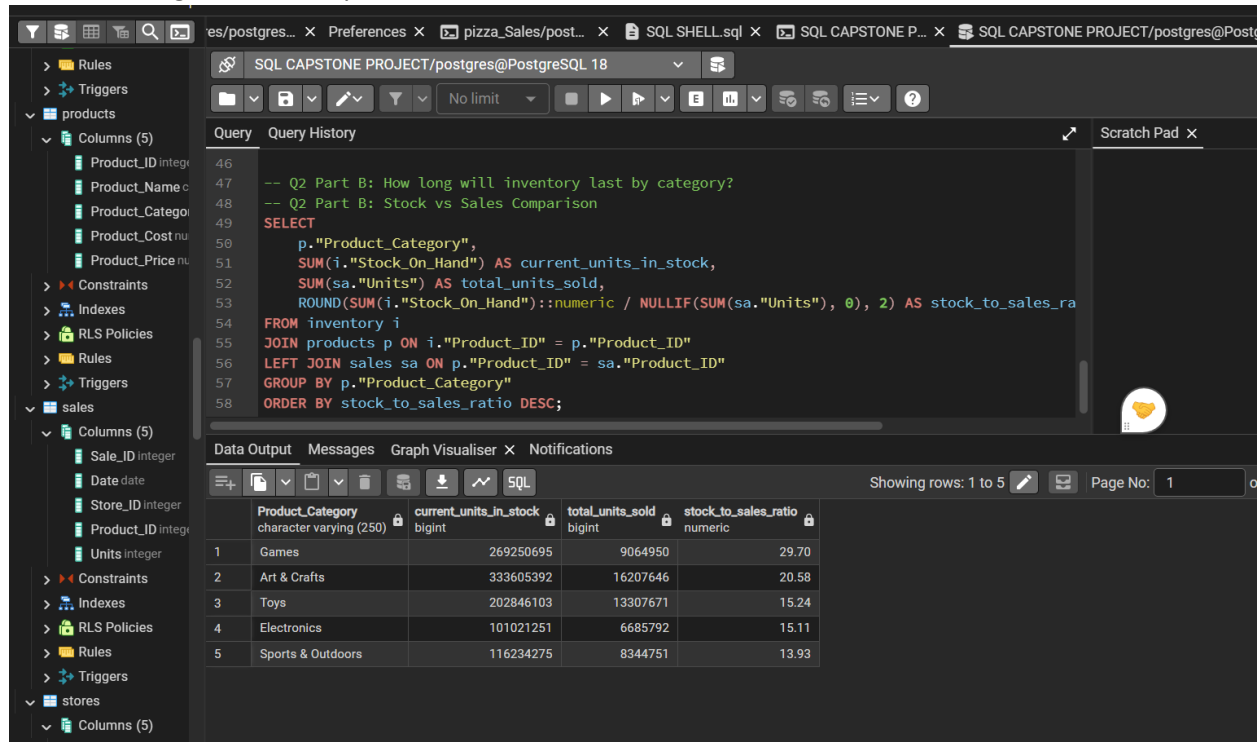
```
24 -- Q2: How much money is tied up in inventory?
25 SELECT
26     p."Product_Category",
27     SUM(i."Stock_On_Hand") AS total_units_in_stock,
28     SUM(i."Stock_On_Hand" * p."Product_Cost") AS total_inventory_value
29 FROM inventory i
30 JOIN products p ON i."Product_ID" = p."Product_ID"
31 GROUP BY p."Product_Category"
32 ORDER BY total_inventory_value DESC;
33
34
35 -- Q1 Part B with Dollar Sign
36 SELECT
```

	Product_Category character varying (250)	total_units_in_stock bigint	total_inventory_value numeric
1	Toys	7553	99861.47
2	Art & Crafts	8635	65075.65
3	Sports & Outdoors	4981	53077.19
4	Games	6155	51489.45
5	Electronics	2418	30705.82

Insight: The most capital is tied up in the Toys category with an inventory value of \$99,861.47 and 71,553 units. Electronics has the lowest inventory value at \$30,705.82. This shows Toys represents the largest inventory investment for Maven Toys.

Recommendation: Maven Toys should closely monitor Toys inventory to avoid overstocking and free up cash. They can also consider increasing Electronics stock since it ties up less capital but may have higher turnover.

Q2B. How long will inventory last



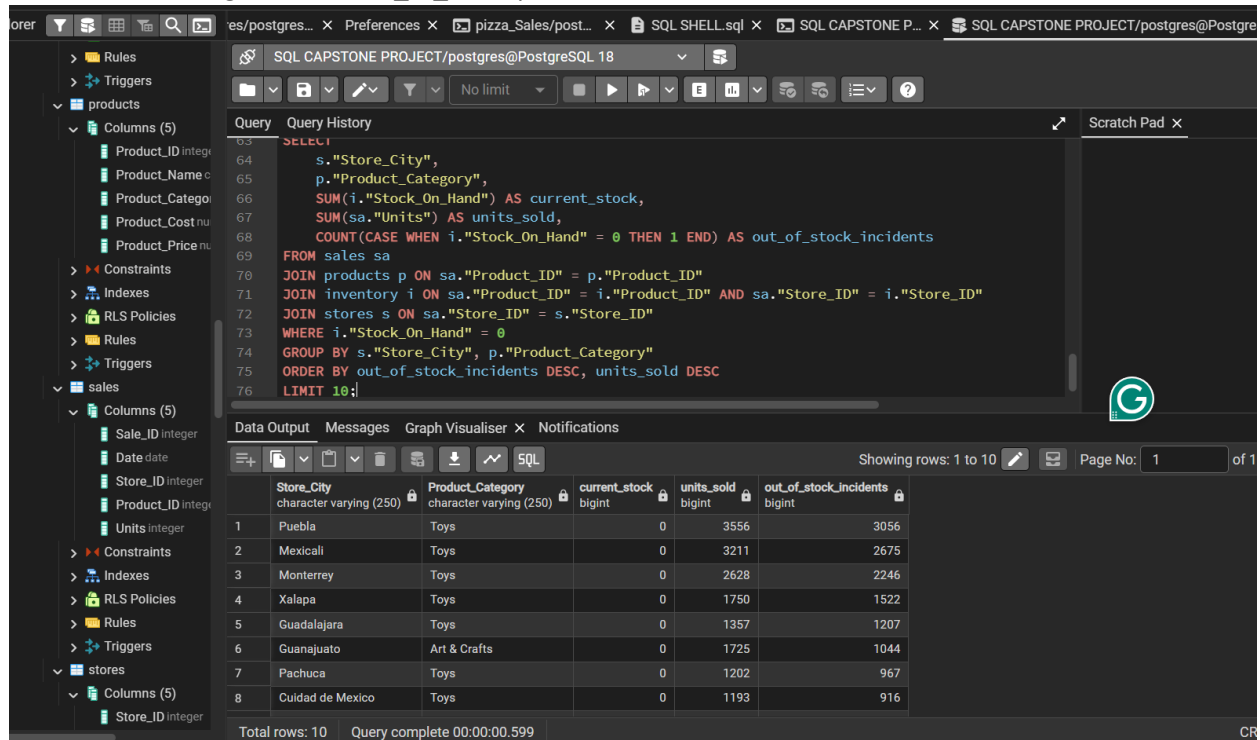
The screenshot shows a PostgreSQL SQL client interface. The left sidebar displays a database schema with tables like 'products', 'sales', and 'stores'. The main window shows a SQL query being executed. The query calculates the stock-to-sales ratio for different product categories. The results are displayed in a table with columns: Product_Category, current_units_in_stock, total_units_sold, and stock_to_sales_ratio. The results show that Games has the highest ratio (29.70), followed by Art & Crafts (20.58), Toys (15.24), Electronics (15.11), and Sports & Outdoors (13.93).

```
-- Q2 Part B: How long will inventory last by category?
-- Q2 Part B: Stock vs Sales Comparison
SELECT
  p."Product_Category",
  SUM(i."Stock_On_Hand") AS current_units_in_stock,
  SUM(sa."Units") AS total_units_sold,
  ROUND(SUM(i."Stock_On_Hand")::numeric / NULLIF(SUM(sa."Units"), 0), 2) AS stock_to_sales_ratio
FROM inventory i
JOIN products p ON i."Product_ID" = p."Product_ID"
LEFT JOIN sales sa ON p."Product_ID" = sa."Product_ID"
GROUP BY p."Product_Category"
ORDER BY stock_to_sales_ratio DESC;
```

	Product_Category	current_units_in_stock	total_units_sold	stock_to_sales_ratio
1	Games	269250695	9064950	29.70
2	Art & Crafts	333605392	16207646	20.58
3	Toys	202846103	13307671	15.24
4	Electronics	101021251	6685792	15.11
5	Sports & Outdoors	116234275	8344751	13.93

Insight: The inventory turnover analysis shows that Games has the highest stock-to-sales ratio of 29.70, meaning current inventory is nearly 30 times total sales. This indicates stock will last excessively long and risks becoming dead inventory. Sports & Outdoors has the lowest ratio of 13.93, meaning inventory turns over faster and may not last long enough to meet demand. Recommendation: Reduce purchasing and run discounts for high-ratio categories like Games and Art & Crafts to clear stock faster. Increase reorder frequency for low-ratio categories to ensure inventory doesn't run out.

Q3. Are sales being lost with out_of_stock products at certain locations



The screenshot shows a PostgreSQL SQL client interface with a query window and a results table. The query is a complex SQL statement that joins the sales, products, inventory, and stores tables to identify out-of-stock incidents. The results table displays the top 10 rows of data, showing the store city, product category, current stock, units sold, and the number of out-of-stock incidents.

```
SELECT
  s."Store_City",
  p."Product_Category",
  SUM(i."Stock_On_Hand") AS current_stock,
  SUM(sa."Units") AS units_sold,
  COUNT(CASE WHEN i."Stock_On_Hand" = 0 THEN 1 END) AS out_of_stock_incidents
FROM sales sa
JOIN products p ON sa."Product_ID" = p."Product_ID"
JOIN inventory i ON sa."Product_ID" = i."Product_ID" AND sa."Store_ID" = i."Store_ID"
JOIN stores s ON sa."Store_ID" = s."Store_ID"
WHERE i."Stock_On_Hand" = 0
GROUP BY s."Store_City", p."Product_Category"
ORDER BY out_of_stock_incidents DESC, units_sold DESC
LIMIT 10;
```

	Store_City character varying (250)	Product_Category character varying (250)	current_stock bigint	units_sold bigint	out_of_stock_incidents bigint
1	Puebla	Toys	0	3556	3056
2	Mexicali	Toys	0	3211	2675
3	Monterrey	Toys	0	2628	2246
4	Xalapa	Toys	0	1750	1522
5	Guadalajara	Toys	0	1357	1207
6	Guanajuato	Art & Crafts	0	1725	1044
7	Pachuca	Toys	0	1202	967
8	Cuidad de Mexico	Toys	0	1193	916

Total rows: 10 Query complete 00:00:00.599

Insight: Puebla had the highest out-of-stock incidents for Toys with 3056 incidents, despite generating 3556 units in sales. Mexicali and Monterrey also had high out-of-stock rates for Toys. This shows significant potential lost sales due to poor inventory replenishment, especially in the Toys category. Recommendation: Maven Toys should implement real-time inventory tracking and automated reorder systems for Toys in top cities like Puebla, Mexicali, and Monterrey. Increasing safety stock for high-demand locations will prevent stockouts and capture lost revenue.